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ORIGINAL ARTICLE

Study on mathematical model of work roll wear in skin-pass rolling of hot steel strip

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Abstract
Skin-pass rolling (or temper rolling) is the final operational step in the production chain of hot-rolled strips, and its main role is to apply a small deformation to the hot-rolled strips cooled to room temperature to improve the shape quality and mechanical properties. Because of the absence of rolling lubricants between the work roll and strip as well as the existence of obvious oxide layer on the strip surface, the wear of work roll is very serious, which exerts an adverse effect on the shape control precision of hot-rolled strips. In this study, the wear law of work roll is obtained by a theoretical analysis of roll wear mechanisms as well as an elastic-plastic finite element analysis of skin-pass rolling. Using the above-mentioned wear law, a work roll wear model is constructed with parameters optimized by the three-population particle swarm optimization and differential evolution algorithm. The newly established wear model is validated experimentally in the field. Calculated values coincide well with those obtained from experiment. The wear model can be used for guiding the actual production.

Keywords Skin-pass rolling of hot steel strip · Wear law of work roll · Finite element analysis · Wear model · Differential evolution · Particle swarm optimization

1 Introduction

Skin-pass rolling (or temper rolling) is the final operational step in the production chain of hot-rolled strips [1]. A slight thickness reduction resulting in an elongation of approximately 1 to 4% is applied to the hot-rolled strips cooled to room temperature to ensure the shape quality and improve the mechanical properties and surface quality to some extent [2]. Because of the absence of rolling lubricants between the work roll and strip as well as the existence of obvious oxide layer on the strip surface, the wear of work roll is very serious. Roll wear not only increases roll consumption but also changes roll gap shape, thereby exerting an adverse effect on the strip shape and the performance of a mill [3, 4]. Thus, it is necessary to make a quantitative calculation of roll wear in skin-pass rolling of hot steel strip.

Many researches have been conducted to analyze roll wear in the rolling processes. Spuzic et al. [5] reviewed the key influential factors of the roll wear in hot rolling processes—roll material properties, process parameters, and roll pass design—and discussed experimental results from a two-disc hot wear rig and a scratch tester. Colas et al. [6] analyzed the condition and mechanism responsible for work roll surface deterioration in hot rolling processes by microstructural observations. Tahir et al. [7] evaluated and compared the differences between prediction model with actual roll wear under the condition of different strip grades and work roll materials. Shang et al. [8] developed a cyclical shifting mode for CVC work roll in the compact strip production (CSP) line to uniform the wear of work roll. Li et al. [9] built a wear prediction model of work roll for non-oriented silicon steel sheets considered lubrication, and studied the effect of roll gap lubrication on the flatness and profile control in asymmetry self-compensating work roll hot rolling processes. Li et al. [10] investigated the measured contour and wear conditions of the roll for skin-pass mill and analyzed the effect of roll contour on strip shape. Liu et al. [11] analyzed the characteristics and wear cause of the roll for hot skin-pass mill and built a work roll wear model by utilizing genetic algorithm. Yang et al. [12] proposed the influential coefficient of the roll wear to analyze

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the uneven wear in hot skin-pass rolling processes and studied the comprehensive effects of roll wear, shifting positions, and bending forces on the strip shape. Xu et al. [13] proposed a wear prediction model of work roll for hot skin-pass mill and analyzed the effect of special wear contours on the shape control characteristics. Zhang et al. [14] studied the uneven wear of the roll for 2250 mm hot skin-pass mill and built a work roll wear model with parameters optimized by utilizing genetic algorithm. All those studies only consider the main influential factors of roll wear and build a semi-empirical and semi-theoretical wear model on account of regression analysis of actual industrial data, whereas the wear law of the roll is not considered.

In this study, a work roll wear model considered the wear law of the roll in skin-pass rolling of hot steel strip is constructed. The three-population particle swarm optimization and differential evolution algorithm (Thr_PSODE) [15–17], which presents obvious advantages in solving multi-peak, multi-variable and nonlinearity, is used to optimize the relevant parameters in the model.

2 Characteristics of roll wear

The main type of mill in skin-pass rolling of hot steel strip is a 4-high mill. It has a variety of shape control means, such as work roll bending and work roll shifting, which greatly improve the shape control capability. Table 1 gives the ranges of process parameters with the 1580 4-high hot skin-pass mill in China as an example.

A lot of work roll contours before and after rolling are measured, and the actual wear contours are obtained by subtracting the roll contours before rolling from the roll contours after rolling (Fig. 1). The wear of work roll exhibits the following characteristics:

(1) Obvious uneven wear is detected at the upper and lower rolls, and the actual roll contours are substantially symmetrical along full roll body.
(2) The axial distance from the wear peak on one side of the roll body to that on the other side is roughly the same as the width of the strip, which indicates that the severe wear area of work roll is in the vicinity of the edge of the strip.
(3) The wear amount of central area is nearly 10 μm, while that of the severe wear area is up to 70 μm. The severe wear area is defined as the wear pit area.
(4) By observing the roll body surface after skin-pass rolling, scattered, line-like pits and bumps, known as orange peel defects (Fig. 2), appear in the dimensional range of 15~210 mm from strip width edge (corresponding to the wear pit area).

Table 1 Ranges of process parameters for the 1580 hot skin-pass mill

Parameters	Value
Rolling force (kN)	2000~4000
Bending force of the work roll (kN)	− 1200~+ 1200
Shifting position of the work roll (mm)	− 200~+ 200
Rolling velocity (m s^{-1})	0~5
Elongation of the strip	1~4%
Changing period of the work roll (kg)	300,000~500,000

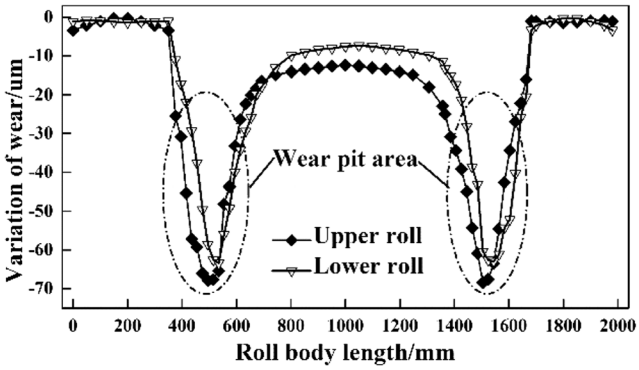


Fig. 1 Actual wear contours of the upper and lower work rolls

3 Wear law of work roll

The wear pit area produced in skin-pass rolling of hot steel strip can affect the contact friction between the work roll and strip, thereby exerting a certain effect on the wear law of work roll.

3.1 Analysis of roll wear mechanisms

In consideration of skin-pass rolling conditions and the above-mentioned roll wear characteristics, the main wear



Fig. 2 Orange peel defects of the roll body surface

mechanisms of work roll are fatigue wear and abrasive wear. The work roll suffers from surface fatigue under alternating stress. The oxide layer debris formed under skin-pass rolling force constantly abrades the roll body, thereby leading to abrasive wear. As both fatigue wear and abrasive wear are involved in stress between the work roll and strip [18], the size of the contact stress directly affects the state of roll wear, and the wear can increase in the high contact stress area.

Taking an example of the upper work roll, a schematic drawing of roll wear mechanisms is obtained, as illustrated in Fig. 3. Wear contours are symmetrical along the roll body; thus, the operating side (OS) is used in the analysis. The wear area of work roll can be divided into two parts along the roll body: the central wear area surrounded by points *O* and *P* and the wear pit area surrounded by points *P*, *Q*, and *R*.

Because of the existence of the wear pit area, the contact stress near point *P* which is in contact with strip increases, namely, the "high-stress area" is produced. In consideration of the above-mentioned wear mechanisms, the size of the contact stress directly affects the wear state, so the wear amount of the "high-stress area" becomes larger, where that of point *P* is the largest. As the wear amount of "high-stress area" increases, point *P* slowly moves towards the central area of roll body, and at the same time, its nearby contours become smooth. Part areas on the left of point *Q* are located in the "high-stress area," which makes the wear amount of these areas larger. Thus, point *Q* slowly moves towards its left-hand side (i.e., the central area of roll body). Compared with that of point *P*, the contact stress of point *Q* is smaller; thus, the movement distance of point *Q* is smaller. Point *R* is outside the range of "high-stress area"; thus, the contact stress of this point does not change significantly, thereby resulting in no locational change.

3.2 Wear law of work roll

In consideration of the above-mentioned roll wear mechanisms and a lot of actual wear contours, the wear law of work roll is revealed, as illustrated in Fig. 4. It mainly includes two stages:

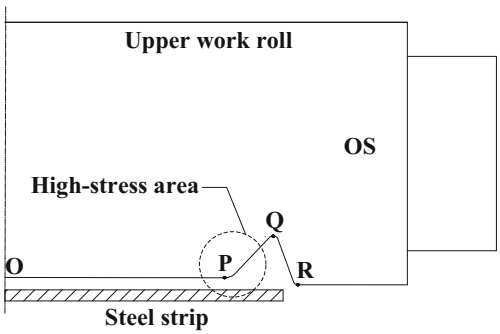


Fig. 3 Schematic drawing of roll wear mechanisms

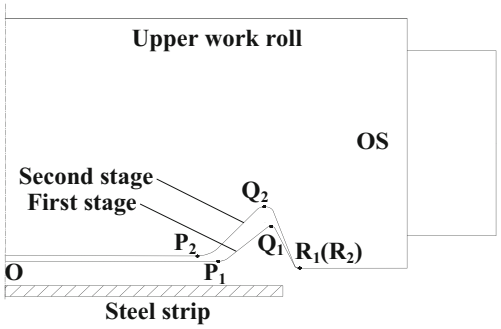


Fig. 4 Schematic drawing of the wear law of work roll

(1) The first stage is the forming process of the wear pit area, and its characteristics are as follows: The roll body which is in contact with strip edge is badly worn, while the central area of the roll body is mildly worn. During skin-pass rolling, the uneven distribution of contact stress occurs under the condition of an initial grinding roll shape. The contact stress on the area which is in contact with strip edge is greater, thereby resulting in the wear aggravation of this area. Finally, the wear contour illustrated in the first stage of Fig. 4 is formed.

(2) The second stage is the developing process of the wear pit area, and its characteristics are as follows: As the skin-pass rolling length increases, the wear increasing speed of the wear pit area is faster than that of the central area. Point P_1 slowly moves towards the central area of roll body and eventually reaches point P_2. Meanwhile, the nearby contours of point P_2 gradually become smooth. Point Q_1 slowly moves towards its left-hand side (i.e., the central area of roll body) and eventually reaches point Q_2. However, compared with that of P_1, the movement distance of point Q_1 is smaller. The location of point R_1 remains unchanged. Finally, the wear contour illustrated in the second stage of Fig. 4 is formed.

4 Finite element analysis of wear law in skin-pass rolling of hot steel strip

To investigate the wear law of work roll, a static implicit simulation model of skin-pass rolling [19–21] is developed utilizing the commercial finite element package ABAQUS.

4.1 Simplification and assumption of hot skin-pass rolling

Hot skin-pass rolling is a complex problem of nonlinearity of material, geometry, and contact. In consideration of the calculation efficiency and precision, the model is simplified and assumed as follows.

(1) On account of the symmetry around the horizontal center line in strip thickness direction, the upper half of the rolls and strip are included in this model. On account of the symmetry around the vertical center line in strip width direction, the right half of the rolls and strip are included in this model.
(2) During actual skin-pass rolling processes, the strip length is considered as infinite. For more accurate simulation of on-line skin-pass rolling processes [22–24], the strip length includes three sections: head stable, skin-pass rolling, and tail stable sections. The head and tail of the strip are extremely long and do not need to be skin-pass rolled. The middle of the strip is the third section, namely, skin-pass rolling section. A schematic drawing of skin-pass rolling state is illustrated in Fig. 5.
(3) The strip is a continuous and homogeneous elastic-plastic material.

4.2 Model parameters settings

Compared with the small reduction condition, the elastic deformation between rolls in skin-pass rolling of hot steel strip cannot be ignored; thus, the rolls are modeled as elastic materials. Taking a 1580 4-high hot skin-pass mill in China as the research object,

the geometrical parameters of model are listed in Table 2. The elasticity modulus of the rolls and strip is 210 and 206 GPa, respectively. Poisson's ratio is set as 0.3 for both the rolls and strip. The work hardening model of the strip is illustrated in Fig. 6.

4.3 Selection of mesh elements

Given that the model involves the contact problem and the contact stress is the variable we need to analyze, the element type of the rolls and strip is set as 3D first-order (linear) hexahedron elements with full integration and incompatible modes (C3D8I in ABAQUS). For the steel strip, the number of the elements in the strip thickness direction is 4, and that in the strip width direction is 240. The dimensions of the elements in the strip length direction are 2.5 mm at the skin-pass rolling section and gradually increase from the central part to the edges at the head and tail stable sections. For the rolls, the elements along the axial direction are 20 mm, and the finer elements along the circumferential direction are used at the contact zones [25]. The mesh elements for the model are illustrated in Fig. 7.

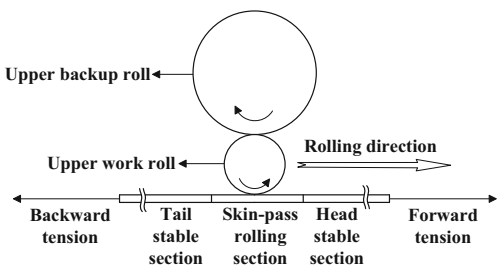


Fig. 5 Schematic drawing of skin-pass rolling state

Table 2 Geometrical parameters of model

Parameters	Value (mm)
Diameter of the work roll body	540
Length of the work roll body	1980
Diameter of the work roll neck	318
Distance between work roll bearings	2740
Diameter of the backup roll body	1080
Length of the backup roll body	1580
Diameter of the backup roll neck	595
Distance between backup roll bearings	2476
Strip width	600
Strip thickness before rolling	2
Length of skin-pass rolling section	150
Length of head stable section	1425
Length of tail stable section	1425

4.4 Contact settings

The frictions between backup roll and work roll and between work roll and strip are modeled by adopting the Coulomb friction model with a friction coefficient of 0.3 [26–28], and the contact problems between backup roll and work roll and between work roll and strip are calculated adopting the penalty method for tangential sliding and normal penetration.

In ABAQUS, rotating a deformable body directly is difficult, while it is relatively easy to rotate a rigid body. Thus, a rigid cell is created at the work roll end on the operating side of

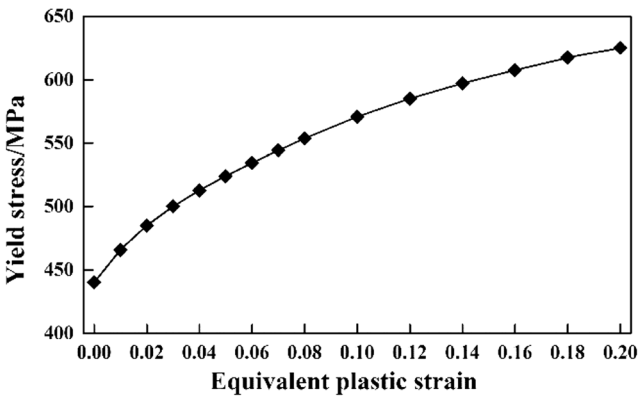


Fig. 6 Work hardening model of strip

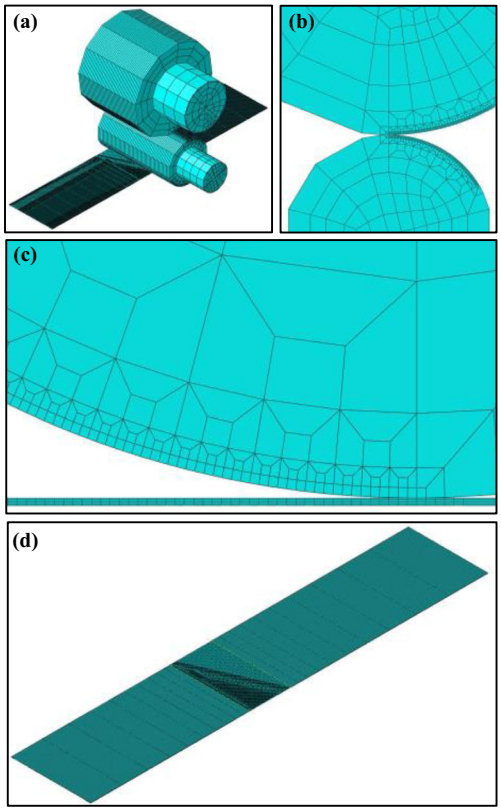


Fig. 7 Mesh elements for the model. **a** Whole structure of the model. **b** Partial enlarged view of the rolls. **c** Partial enlarged view of the deformation zone. **d** Partial enlarged view of the strip

the mill. The rigid cell and work roll are firmly bonded together by adopting Tie constraint, and the rotation of the reference point on the rigid cell is defined to rotate the work roll.

4.5 Boundary condition and loadings settings

The coordinates in the width, thickness, and length directions of the strip are x, y, and z, respectively. In order to ensure the displacement freedom of the rolls in the y direction and the rotation freedom of each roll in the x direction in the model, the model controls the freedom of each roll by controlling the freedom of the rigid cells tied to each roll end. The displacement in the x direction and rotations in the y and z directions of the strip left surface with half width are constrained by symmetry, and the displacement in the y direction and rotations in the x and z directions of the strip bottom surface with half thickness are constrained by symmetry. The backward and forward tensions are imposed on the tail and head of the strip, respectively. The bending forces in the y direction are imposed on the rigid cells tied to work roll end.

A certain vertical downward displacement of the rigid cell tied to backup roll end is applied to conduct screwing down. Thereafter, the rolls are rotated by rotating the rigid cells tied to work roll end, thereby completing the simulation of skin-pass rolling.

5 Results and discussion

In consideration of the above-mentioned roll wear mechanisms, the size of contact stress directly affects the state of roll wear; thus, the contact stress is analyzed and contrasted.

The simulation conditions of the model are listed in Table 3. When work roll shape is set as an initial grinding contour or the contours corresponding to the first and second wear stages (Fig. 8), the contact stress between the strip and work roll (hereinafter referred to as "the contact stress") is calculated, as illustrated in Fig. 9. The calculation results are as follows:

1. When work roll shape is an initial grinding contour, the uneven distribution of contact stress along the roll body occurs, and the contact stress on the area which is in contact with the strip width edge is greater.
2. Compared with that of an initial grinding contour, the contact stress of the contours corresponding to the first and second wear stages presents the following characteristics:

   (a) The location of maximum contact stress moves from the edge of strip width towards the central area of roll body. The locations of the maximum contact stress corresponding to the first and second wear stages coincide with those of points P_1 and P_2 in Fig. 8, respectively. A "high-stress area" exists in the vicinity of the maximum contact stress.
   (b) The contact stress in the vicinity of the strip width edge changes slightly, but the contact stress on its left side increases.
   (c) The contact stress in the central area of roll body slightly decreases.

Thus, when work roll shape is the contours corresponding to the first and second wear stages, the contact stress distribution coincides well with the obtained roll wear mechanisms, and the correctness of the wear law is proved.

Table 3 Simulation conditions of the model

Parameters	Value
Thickness reduction ratio of the strip	2.5%
Bending force of the work roll (kN)	0
Shifting position of the work roll (mm)	0
Forward tension (MPa)	110
Backward tension (MPa)	45

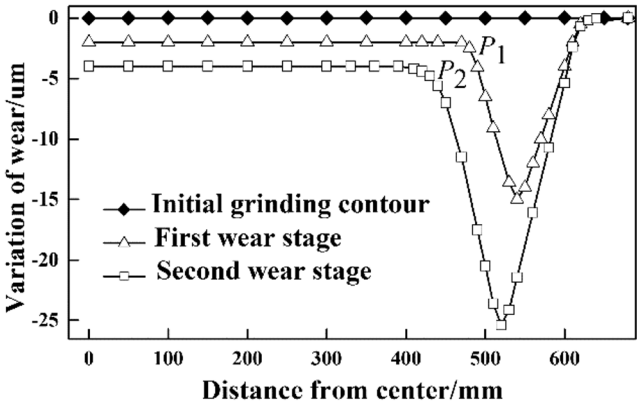


Fig. 8 Loaded work roll shapes corresponding to different stages

6 Wear model based on the wear law of work roll

A wear model in skin-pass rolling of hot steel strip, which meets actual wear condition, is constructed on the basis of the obtained wear law of work roll.

After the rolling of i coils, the wear area of work roll can be divided into four parts along the roll body: the center wear area surrounded by points O and P, edge I wear area surrounded by points P and Q, edge II wear area surrounded by points Q and S, and edge III wear area surrounded by points S and R (Fig. 10). Variables k_1 and k_2 are the ratio of wear amount of points Q and S relative to that of the center wear area. Variables d_0^i, d_1^i, d_2, and d_3 are the distance between points P, Q, S, and R and the strip width edge, respectively; among these variables, d_0^i and d_1^i are the ones associated with the wear law.

In consideration of the above-mentioned wear law, variables d_0^i and d_1^i gradually increases as the skin-pass rolling length increases. However, the increasing speed of d_1^i is slower than that of d_0^i, and variables d_2 and d_3 remain unchanged. The roll wear mechanisms show that variables d_0^i and d_1^i are affected mainly by skin-pass rolling force and skin-pass rolling length. These variables can be expressed as

$$d_0^i = k_0^l\left(l_r^i/D_w\right) + k_0^p p_r^i + c_0 \tag{1}$$

$$d_1^i = k_1^l\left(l_r^i/D_w\right) + k_1^p p_r^i + c_1 \tag{2}$$

where k_0^l is the influence coefficient of skin-pass rolling length corresponding to variable d_0^i (mm); l_r^i is the accumulated length after the rolling of i coils (m); D_w is the actual work roll diameter (mm); k_0^p is the influence coefficient of skin-pass rolling force corresponding to variable d_0^i (mm kN^{-1}); p_r^i is the accumulated skin-pass rolling force after the rolling of i coils (kN); c_0 is the constant value corresponding to variable d_0^i (mm); k_1^l is the influence coefficient of skin-pass rolling length corresponding to variable d_1^i (mm); k_1^p is the influence coefficient of skin-pass rolling force corresponding to variable d_1^i (mm kN^{-1}); and c_1 is the constant value corresponding to variable d_1^i (mm).

Figure 11 shows a common discrete slice method used in the wear model. The work roll is evenly divided into s slices from the driving side (DS) to the operating side (OS). After each strip is rolled, the wear amount of each slice is calculated on the basis of the actual process parameters. After a certain number of strips are rolled, the wear amount of each slice is accumulated, and the overall axial distribution curve of the roll body wear is finally obtained. In the figure, x_j is the coordinate of slice j relative to skin-pass rolling centerline ($x = 0$); n_l is the slice number corresponding to the strip width edge of DS; n_r is the slice number corresponding to the strip width edge of OS; B is the strip width (mm); S_i is the work roll

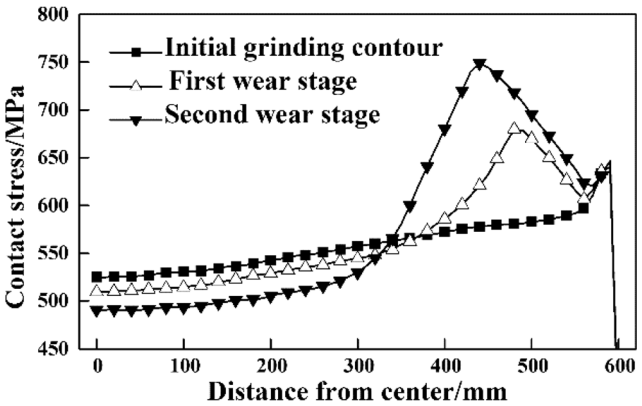


Fig. 9 Contact stress of the work roll surface corresponding to different stages

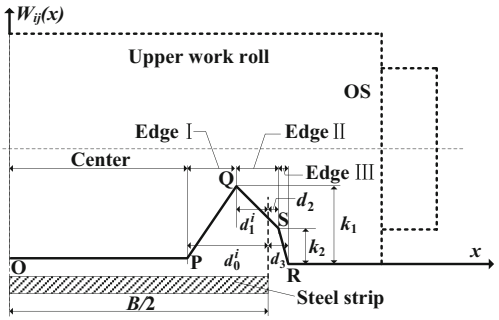


Fig. 10 Axial distribution characteristics of roll wear after the rolling of i coils

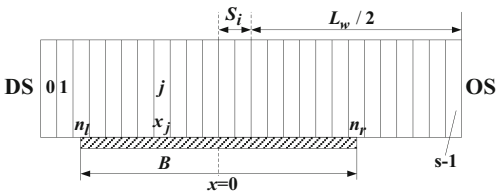


Fig. 11 Discrete slice method along the roll body

shifting position corresponding to strip i (mm); and L_w is the length of work roll body (mm).

After strip i is rolled, the wear amount of slice j can be expressed as

$$w_{ij} = k(1 + k_d(D_{w0} - D_w))e^{\left(-l_r^i/w_{sh}\right)} \frac{l_s^i}{\pi D_w(1 + f_s)} p^i f\left(B, x_j\right) \tag{3}$$

where k is the basic wear rate ($mm^2\ kN^{-1}$); k_d is the influence coefficient of roll wear diameter (mm^{-1}); D_{w0} is the work roll diameter before rolling (mm); w_{sh} is the coefficient of surface hardening (m); l_s^i is the actual length of strip i (m); f_s is the forward slip factor; p^i is the unit width rolling force corresponding to strip i ($kN\ mm^{-1}$); and $f(B, x_j)$ is the axial distribution function of uneven wear. The $f(B, x_j)$ on the OS can be expressed as

(1) Center area:

$$f\left(B, x_j\right) = 1, 0 \le x_j < \frac{B}{2} - d_0^i \tag{4}$$

(2) Edge I area:

$$f\left(B, x_j\right) = \frac{k_1 - 1}{d_0^i - d_1^i}\left(x - \frac{B}{2} + d_0^i\right) + 1, \frac{B}{2} - d_0^i \le x_j < \frac{B}{2} - d_1^i \tag{5}$$

(3) Edge II area:

$$f\left(B, x_j\right) = \frac{k_2 - k_1}{d_1^i + d_2}\left(x_j - \frac{B}{2} + d_1^i\right) + k_1, \frac{B}{2} - d_1^i \le x_j < \frac{B}{2} + d_2 \tag{6}$$

(4) Edge III area:

$$f\left(B, x_j\right) = \frac{-k_2}{d_3 - d_2}\left(x_j - \frac{B}{2} - d_2\right) + k_2, \frac{B}{2} + d_2 \le x_j \le \frac{B}{2} + d_3 \tag{7}$$

Given that the wear contours are symmetrical along the roll body, the expression of the DS cannot be repeated.

In Eqs. (4)–(7), x_j can be expressed as

$$x_j = jd_x - L_w/2 + S_i \tag{8}$$

where d_x is the slice width (mm).

After strip k is rolled, the accumulated wear amount of slice j can be expressed as

$$U_w(j) = \sum_{i=1}^{k} w_{ij}(j)\ ,\quad 1 \le k \le n\ ,\quad 0 \le j \le s-1 \tag{9}$$

where $w_{ij}(j)$ is the wear amount produced by strip i (mm); n is the total number of strips in a rolling schedule; and s is the total number of slices along the roll body.

After the wear model is established, the next crucial issue is to solve model parameters. The wear model determined by Eqs. (1)–(9) contains three types of parameters. The first type is the basic skin-pass rolling process parameters, including D_{w0}, D_w, L_w, l_r^i, p_r^i, l_s^i, f_s, p^i, B, d_x, and S_i. The second type is the uneven wear coefficients obtained by measuring a lot of wear contours by means of a high-precision grinding machine, including d_2, d_3, k_1, and k_2, as listed in Table 4. The third type is the model parameters to be optimized, including k, k_d, w_{sh}, k_0^l, k_0^p, c_0, k_1^l, k_1^p, and c_1.

7 Three-population particle swarm optimization and differential evolution algorithm

Three-population particle swarm optimization and differential evolution algorithm (Thr_PSODE), which effectively combines the local search capability of particle swarm optimization (PSO) [29–31], the global convergence property of differential evolution (DE) [32–34], and the characteristics of different evolution modes of DE [35, 36], is a hybrid optimization algorithm with better efficiency and precision. In Thr_PSODE, two populations are used for DE optimization of best and rand evolution modes respectively, and the remaining one is used for PSO optimization.

Thr_PSODE includes the following calculation steps:

(1) Initialize the following parameters of Thr_PSODE: the population scale, dimensions of each particle or

Table 4 Uneven wear parameters in roll axial direction

Parameters	d_2 (mm)	d_3 (mm)	k_1	k_2
Value	0	35	7.72	2.49

individual, maximum iteration times, mutation factor and crossover constant of DE, and cognitive and social factors of PSO.

(2) Initialize the entire population. The entire population is divided into three subpopulations: subpopulations 1 and 2 are used for DE optimization with position initialization of each individual, and subpopulation 3 is used for PSO optimization with position and velocity initialization of each particle.

(3) Preliminarily select the global best particle or individual among three subpopulations. Calculate the fitness of particles or individuals of each subpopulation respectively, and select the global best fitness (F^1_{gbest}, F^2_{gbest}, F^3_{gbest}) and the corresponding particle or individual. The three global best fitness values are compared, and the minimum fitness (F^{min}_{gbest}) is obtained. If $F^{min}_{gbest} = F^1_{gbest}$, F^2_{gbest} and F^3_{gbest} should be replaced with F^1_{gbest}. Besides, the global best positions of subpopulations 2 and 3 should be replaced with the global best ones of subpopulation 1, and the velocity of the global best particle of subpopulation 3 should be set to 0. If $F^{min}_{gbest} = F^2_{gbest}$, the processing method is similar. If $F^{min}_{gbest} = F^3_{gbest}$, the positions and fitness values of the global best individual of subpopulations 1 and 2 should be replaced with the one of the global best particle of subpopulation 3.

(4) Execute mutation, crossover, and selection operations for subpopulations 1 and 2, and update position and velocity for subpopulation 3. DE has different differential evolution modes, in which subpopulations 1 and 2 adopt rand and best mutation modes respectively. For subpopulation 1, in rand mutation mode, the new individuals are generated using Eq. (10):

$$V_i(t) = X_{r1}(t) + F(X_{r2}(t) - X_{r3}(t)) \tag{10}$$

where i is the sequence number of individuals; $i \in [1, 2, \ldots, 3, N]$; N is the total number of individuals, i.e., the population scale; t is the iteration times; $r1 \neq r2 \neq r3 \neq i$ and $r1$, $r2$, $r3 \in [1, N]$; $F \in [0, 1]$ is the mutation factor; X_{r1} is the basic vector; and $X_{r2} - X_{r3}$ is the difference vector of individuals.

For subpopulation 2, in best mutation mode, the new individuals are generated using Eq. (11):

$$V_i(t) = X_{best}(t) + F(X_{r1}(t) - X_{r2}(t)) \tag{11}$$

where X_{best} is the best individual of the current generation.

For subpopulations 1 and 2, in the crossover operation, the individuals of a new trial population $U_i(t) = [u_{i1}, u_{i2}, \ldots, u_{iD}]$ are generated by the mutation individuals $V_i(t) = [v_{i1}, v_{i2}, \ldots, v_{iD}]$ and the target individuals $X_i(t) = [x_{i1}, v_{i2}, \ldots, x_{iD}]$ in the following way:

$$u_{ij} = \begin{cases} v_{ij}, \text{if}(randb(j) \le CR) \text{ or } j = ranbr(i) \\ x_{ij}, \text{otherwise} \end{cases} \tag{12}$$

where $j \in [1, D]$, D is the dimensions of each individual; $randb(j) \in [0, 1]$; $CR \in [0, 1]$ is the crossover constant; $ranbr(i) \in [1, 2, \ldots, D]$ is a randomly selected integer number.

For subpopulations 1 and 2, in the selection operation, the fitness of crossover individuals is evaluated, and the new individuals are generated using Eq. (13):

$$X_i(t+1) = \begin{cases} X_i(t), \text{if}(f(X_i(t)) \le f(U_i(t))) \\ U_i(t), \text{otherwise} \end{cases} \tag{13}$$

For subpopulation 3, the position and velocity of each particle are updated using Eqs. (14) and (15):

$$v_i(t+1) = k \cdot (w \cdot v_i(t) + c_1 r_1 (pbest_i(t) - x_i(t)) + c_2 r_2 (gbest_i(t) - x_i(t))) \tag{14}$$

$$x_i(t+1) = x_i(t) + v_i(t+1) \tag{15}$$

where c_1 is the cognitive factor; c_2 is the social factor; r_1, $r_2 \in [0, 1]$; $pbest_i(t)$ is the best position of each individual; $gbest_i(t)$ is the global best position of the population.

w is the inertia weight, which can be calculated using Eq. (16):

$$w(t) = 0.9 - \frac{t}{t_{\max}} \times 0.5 \tag{16}$$

where $t_{\max}$ is the maximum iteration times.

k is the constriction factor adopted to improve the global convergence property:

$$k = \frac{2}{\left|2 - l - \sqrt{l^2 - 4l}\right|}, l = c_1 + c_2, l > 4 \tag{17}$$

(5) Judge the termination conditions. If the termination conditions of iteration are not met, then return to Step (4); otherwise, end the optimization processes.

8 Parameter optimization of wear model

8.1 Establishment of optimization target

The parameter optimization of wear model aims to obtain the smallest deviation between the calculated and actual wear values of each slice along the roll body. During skin-pass rolling of hot steel strip, the work roll changing period varies

widely and randomly because of all sorts of surface defects; thus, the actual wear data corresponding to different roll changing periods is partially available. Meanwhile, the corresponding calculation values can be obtained by the newly established wear model. The optimization equation can be expressed as

$$\min f = \sqrt{\sum_{i=1}^{b}\sum_{j=0}^{s-1}\left(\frac{w_{ij}^{c}-w_{ij}^{m}}{w_{ij}^{m}}\right)^{2}} \tag{18}$$

where w_{ij}^{c} and w_{ij}^{m} are the calculated and actual wear values of slice j of the ith work roll after rolling in a rolling schedule, respectively, and b is the number of work roll after rolling in a rolling schedule, $1 \le b \le n$.

8.2 Constraints of optimization

The parameter optimization of wear model need to be set in accordance with the following constraints:

(1) The constraints of d_0^i and d_1^i. According to wear law, d_0^i gradually increases, but is bound by the strip width, i.e., $d_0^i \le B/2$. Moreover, the increase in d_1^i is bound by d_0^i, i.e., $d_1^i \le d_0^i$. Therefore,

$$d_1^i \le d_0^i \le B/2 \tag{19}$$

(2) The constrain of work roll material, which is expressed as

$$\begin{cases} k_{\min} \le k \le k_{\max} \\ k_{d,\min} \le k_d \le k_{d,\max} \\ w_{sh,\min} \le w_{sh} \le w_{sh,\max} \end{cases} \tag{20}$$

where k, $k_{\min}$, and $k_{\max}$ are the basic wear rate and the minimum and maximum values of the basic wear rate, respectively; k_d, $k_{d,\min}$, and $k_{d,\max}$ are the influence coefficient of roll wear diameter and the minimum and maximum values of the influence coefficient of roll wear diameter, respectively; w_{sh}, $w_{sh,\min}$, and $w_{sh,\max}$ are the coefficient of surface hardening and the minimum and maximum values of the coefficient of surface hardening, respectively.

8.3 Parameter optimization of wear model based on Thr_PSODE

The wear model for hot skin-pass mill presents a multi-peak, multi-variable, and highly nonlinear parameter optimization problem. Thus, Thr_PSODE is adopted in this study because of its overall superiority in the local search capability and global convergence properties.

Parameter optimization of wear model based on Thr_PSODE includes the following calculation steps:

(1) Initialize the skin-pass rolling parameters: (1) initial parameters, including the geometrical parameters of the strips and rolls, and the rolling conditions parameters; (2) the ranges of constraints, including constraints associated with the wear law, and roll material constraints.

Taking a 1580 4-high hot skin-pass mill in China as the research object, the geometrical parameters of the strips and rolls, the rolling conditions parameters can refer to Tables 2 and 3, respectively. The setting method of constraints can refer to Section 8.2.

(2) Initialize the following parameters of Thr_PSODE: the population scale, dimensions of each individual or particle, maximum iteration times, mutation factor and crossover constant of DE, and cognitive and social factors of PSO are set to 50, 9, 300, 0.7, 0.4, 2.05, and 2.05, respectively.
(3) Initialize the entire population. The entire population is divided into three subpopulations: Subpopulations 1 and 2 are used for DE optimization with position initialization of each individual and subpopulation 3 is used for PSO optimization with position and velocity initialization of each particle.
(4) Verify the validity of each individual or particle. Use Eqs. (1) and (2) to calculate variables d_0^i and d_1^i in accordance with the parameters of each individual or particle. Then, discarded and regenerated the individual or particle that does not meet Eq. (19).
(5) Preliminarily select the global best particle or individual among three subpopulations. Calculate the fitness of

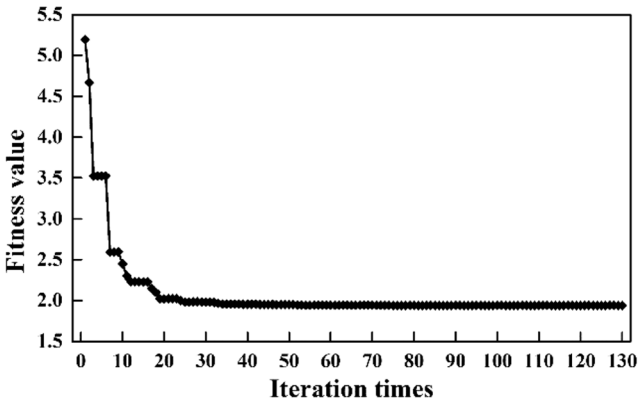


Fig. 12 Target evolution curve of Thr_PSODE

Table 5 Optimized parameters of wear model

Parameters	Value
K (mm^2 kN^{-1})	1.034×10^{-6}
k_d (mm^{-1})	4.703×10^{-3}
w_{sh} (m)	4.998×10^{6}
k_0^l	2.559×10^{-3}
k_0^p (kN^{-1})	2.507×10^{-3}
c_0	1.243×10^{2}
k_1^l	2.394×10^{-3}
k_1^p (kN^{-1})	1.926×10^{-3}
c_1	4.613×10^{1}

particles or individuals of each subpopulation respectively, and select the global best fitness and the corresponding particle or individual. The three subpopulations should be compared and substituted to determine the global best fitness and the corresponding particle or individual.

(6) Execute mutation, crossover, and selection operations for subpopulations 1 and 2 and update position and velocity for subpopulation 3.

(7) Judge the termination conditions. If the termination conditions of iteration are not met, then return to step (6); otherwise, end the optimization processes.

(8) Output of the optimization result.

Combined with the above-mentioned optimization process, the target evolution curve of Thr_PSODE is illustrated in Fig. 12. After around 80 generations of evolution, the fitness of the population reaches a stable state. The global best individual at 130th generation is taken as the optimal solution for the model optimization and listed in Table 5.

9 Field application of work roll wear model

The research results of the wear model are applied to a 1580 4-high hot skin-pass mill in China. The wear contours of the work roll corresponding to container plates with a strip width of 1200 mm, a strip thickness of 2 mm, and the roll changing period of 5000, 6000, and 10,000 m are chosen from a lot of actual wear contours. Meanwhile, the corresponding calculation values are obtained using the optimized wear model parameters. Figure 13 shows the actual wear law and a comparison of calculated and actual wear values of work roll. The actual wear law is as follow: As the skin-pass rolling length increases, the wear increasing speed of the wear pit area is faster than that of the central area. Point P slowly moves towards the central area of roll body, and at the same time, its nearby contours become smooth. Point Q slowly moves towards its left-hand side (i.e., the central area of roll body). However, compared with that of point P, the movement distance of point Q is smaller. The position of point R remains unchanged.

The actual wear values coincide well with the calculated values, indicating that the wear model for hot skin-pass mill obtains higher calculation accuracy and can be used for guiding the actual production.

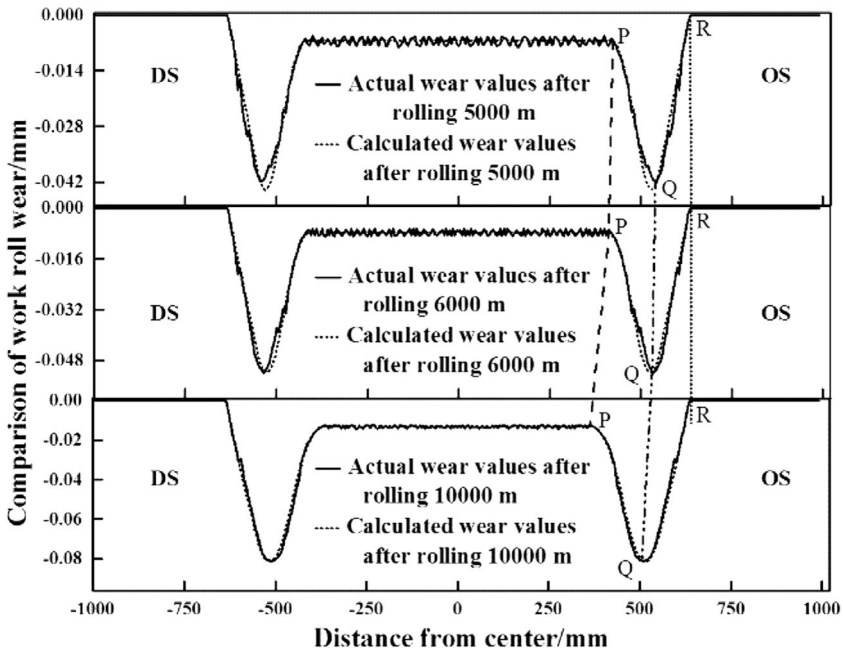


Fig. 13 Comparison of calculated and actual wear values of work roll

10 Conclusion

(1) The wear law of work roll is obtained by a theoretical analysis of roll wear mechanisms as well as an elastic-plastic finite element analysis of skin-pass rolling, as illustrated in Fig. 4: Point P_1 slowly moves towards the central area of roll body, and eventually reaches point P_2. Meanwhile, the nearby contours of point P_2 gradually become smooth. Point Q_1 slowly moves towards its left-hand side (i.e., the central area of roll body) and eventually reaches point Q_2. However, compared with that of P_1, the movement distance of point Q_1 is smaller. The location of point R_1 remains unchanged.

(2) The wear law is quantitatively expressed, and a wear model is constructed. Thr_PSODE, which exhibits overall superiority in the local search capability and global convergence properties, is adopted to optimize the model parameters.

(3) The newly established wear model is validated experimentally in the field. Calculated values coincide well with those obtained from experiment. The model can be used for guiding the actual production.

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